Daniel Mandragona

ML Performance Engineer | JAX & Pallas TPU Kernels · Distributed LLM Training · C++ Systems

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MLE with six years at Google, now focused on large-scale LLM training performance (JAX/Pallas on Cloud TPU v7x). Prior experience building secure and reliable C++ systems. Part-time efforts in QEC research engineering.

TECHNICAL SKILLS

Languages & Frameworks: C++ · Python · Go · C · CUDA · JAX · Pallas · XLA / HLO · MaxText
ML Systems: TPU Kernels · Distributed Sharding & Collectives · FP8 & MoE Training · XProf · Roofline Analysis · QEC

EXPERIENCE

Google
Sep 2018 – Aug 2022; Sep 2024 – Present

Machine Learning Engineer — *Mountain View, CA*

JAX/TPU Performance Engineering — MaxText / JAX Models & Performance
Apr 2026 – Present

DeepSeek-V3 671B Pretraining — MLPerf 6.1

- Led the design and realization of an experimental batch-split pipelined DeepSeek-V3 implementation into a fully MLPerf-compliant submission at 8,192-chip scale.
- Stabilized MoE training: analyzed expert-load imbalance across all 58 sparse layers, reconciled subtle aux-LB loss & dynamic router-bias differences, restored exact FP32 router biases, and debugged a critical dataset issue resulting in faster convergence (53 → 43 steps) and +15% E2E tok/s.
- Overlapped shared experts with SparseCore combine reduce-scatters; coalesced and pipelined cross-slice (DCN) gradient all-reduces across the expert axis (+11% E2E tok/s at 8,192-chip scale).
- Drove the move from multi-slice 4×4×N to single-slice 8×A×B meshes: remapped 3D collectives, fit expert FSDP shards to whole FP8 tiles, and cut 18 relayout copies/step.

TPU Kernel Engineering (Pallas + C++)

- Upstreamed MaxText's mHC-lite Pallas kernel: VMEM-fused norm/projection/gating, closed-form RMSNorm+GEMM VJP, packed MXU contractions, and 3-path FP32 dX accumulation; up to 2.6× vs. pure JAX, +15% E2E tok/s on a customer LLM.
- Built an E2E hybrid-precision Pallas S5 SSM kernel for Oxford FLAIR's LOBS5 limit-order-book model: 2× faster training step than pure-JAX associative scan at 12k sequence length.
- Fused VMEM dQ accumulation and causal row-band skip into Tokamax Splash Attention bwd (−3.3% DSv3 step time).
- Advised on a chunked Gated DeltaNet Pallas kernel (2.2× layer step); derived the closed-form backward gradient-flow graph used to verify numerics across the chunked recurrence.
- Eliminated row-max reductions in FP8 Splash Attention via fixed-ceiling softmax and a fused dynamic norm-bound gate: −22% fwd kernel at 4k, −44% at 128k, and −12–16% step time at 128k–256k without underflow.
- Hand-scheduled a bare-metal C++ fused matmul-chain TPU kernel to 1.3× XLA using a Lean 4 tick model + A* search.
- Found a mathematical trick enabling a 25% speedup on a Pallas FFT kernel for TPU.

Distributed Training & Customer Engagements

- Engineered async host offloading for hybrid-attention scans in MaxText: +15% tok/s, −20% peak HBM on a customer LLM.
- Cut a flagship customer's MoE proxy step time 51% (451 → 223 ms) via Ring-of-Experts EP, Pallas ragged sort, GMM v2, projection fusion, and 2-chunk comm/compute overlap.
- Reduced Wan 2.1 (14B) diffusion step time 25% via Unified Sequence Parallelism (Ring Attention + DeepSpeed-Ulysses).

C++ Performance Engineering — Tesseract Decoder (Quantum AI, 20%)
2025 – Present

- Achieved 5.6× speedup on the d=23 surface code through C++ code optimizations: cutting vector copies with linked lists, utilizing an arena allocator, and compiler prefetching hints.
- Researched accelerator-native QEC decoding for both TPUs & GPUs; built a CUDA implementation for C++ reference decoder alongside TPU version and benchmarked on NVIDIA L4 vs. TPU v5e.
- Upstreamed three merged PRs: C++ hot-path changes, pybind11 support for multiple detector error models, and generated Python type stubs.

Certificate Authority Infrastructure — Google Trust Services
Aug 2019 – Aug 2022; Sep 2024 – Apr 2026

- Eliminated low-level CA bottlenecks across Go/C interop, DB hotspot thrashing, network egress, and DNS resolver latency.
- Designed load balancing for atypical network constraints; migrated C++ servers into hardened, high-security machine pools.
- Implemented ACME (RFC 8555), hardened the C++ CA for public launch; supported WebTrust audits and key ceremonies.
- Discovered and reported a high-severity DoS vulnerability (CVE-2019-17596) in Go's crypto/dsa package.

Google Research — Perception & Play Abuse
2018 – 2019

- Built ML pipelines for image-saliency prediction (ResNet U-Net) and C++ multimodal Google Play abuse detection.

EDUCATION

M.S. Mathematics (GPA 3.93) · Texas A&M University, College Station, TX — Aug 2022 – Dec 2024
B.S. Mathematics, Computational Sciences (GPA 3.71) · University of Central Florida, Orlando, FL — Aug 2013 – May 2018